One protein, across samples

The same values, joined, sample by sample with the conditions in metadata order.

AT A GLANCE

Where	Single protein → Lineplot
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and at least one feature selected.
Data source	Data, Filtered Data or Imputed Data; the page opens on the most processed one that exists.
One point	One feature in one *sample*. The axis is the samples, not the conditions: nothing is averaged.
Several features	One line each, in the colours of the tab10 cycle.
Table	Yes, below the plot, downloadable as CSV — one row per feature and sample.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The boxplot summarises a condition; this page refuses to. Every marker is one sample, and the line joins them with the selected conditions in metadata order, so what the figure shows is the individual runs in sequence: which replicate sits high, whether a condition moves as a block, and where a feature simply was not measured. It is also the only single-feature plot that draws more than one feature at once, which is how two phosphosites on the same protein are compared.

The x axis is an ordering, not a scale. The samples are equally spaced because they are drawn in a list, and the gaps between conditions carry no quantity — a doubling of dose and the next time point look alike here. The line between two markers is drawn to make the sequence readable, not because anything was measured between them.



Figure 1.1. Two proteins, 1433B_HUMAN and 1433E_HUMAN, across the ten samples of the example dataset: the five WT runs first, then the five GRK ones, in metadata order. Each marker is one sample and each line one feature, coloured from the fixed set rather than from the condition palette. Both features are measured in every sample, so neither line has a gap. They run roughly a $\log_2$ unit apart; 1433B_HUMAN is flat across all ten runs, while 1433E_HUMAN sits a little higher in GRK than in WT — a shift of about a tenth of a $\log_2$ unit, which the y range makes visible and which is exactly the kind of step this plot is read for.

Reading it

- One line, one feature. Its colour comes from the fixed tab10 set, in the order the features were selected; it is not the condition palette, because a line crosses every condition.
- One marker, one sample: the value of that feature in that run.
- The spread within a condition is the scatter of the markers standing over consecutive labels of the same condition. There is no error bar — nothing here is a mean.
- The label under each marker: condition_n, numbered within the condition rather than across the experiment, with the sample's own identifier on a second line when **Toggle ID** is on.

> **EASY TO GET WRONG**
>
> A missing value is not drawn, but the line is: the markers on either side of it are joined directly, so a feature missing in the middle of a condition leaves a segment that looks like a measurement and is not. The gap is visible only as a missing marker over a label that is still on the axis. When a trajectory matters, count the markers against the labels, or read the table, which retains missing feature–sample combinations as

NA.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset the features are taken from: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered.	Protein	page
Data source	Which matrix the values are read from: Data, Filtered Data or Imputed Data. Only the ones that exist are offered.	the most processed matrix available	page
Identifier	Which name the features are chosen by: Protein Names, Phosphosite Names or Phosphoprotein Names according to the level, and Gene Names where the dataset carries a gene column. The phosphosite level offers no gene option.	Protein Names	page
Proteins / Sites	The features to draw, one line each. Labelled **Genes** under Gene Names.	the first three in the list	page
Conditions	Which conditions contribute samples. They are listed, and laid out on the x axis, in metadata order; selecting them in another order does not reorder the axis. Within a condition the samples follow the annotation.	all	page
Toggle ID	Adds the sample's own identifier under each condition_n label, on a second line.	off	page
Show NA as low	Draws the line down to 0 on the log2 axis (equivalent to linear 1) at a missing sample, without a marker. This changes only the drawing, not data or statistics.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20, 10, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Show title	The plot's own title is Protein expression across samples at protein level, Gene … under Gene Names and Phosphosite … at both phospho levels. Its text, size, weight, colour and position are set here too.	shown	appearance
Show legend	The legend of features beside the plot, headed Feature, and its title, size, colour and frame.	shown	appearance
Frame, grid, font, background	Which of the four spines are drawn and the grid behind the lines; one face for the whole figure or one per text; paper colour, transparency and the preset themes.	all four spines, grid on both axes	appearance
Condition names	Offered, but this figure builds its sample labels from the annotation's own condition names and does not read the overrides. There is no condition-colour card at all: the lines are coloured by feature.	the annotation's names	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. **Toggle Header** and **Toggle Legend** are no longer beside the plot: they are **Show title** and **Show legend** in the panel.

The panel shows neither a **Points** card nor the condition-renaming rows on this page: the line plot draws a fixed circle marker, and the sample labels are built from the annotation, so neither would have any effect.

3 How it is computed

HOW IT IS COMPUTED

The rows of the chosen matrix whose identifier column matches any selected feature are taken: at protein level the match is on the first accession of the group, by gene on any of a row's semicolon-separated gene tokens.
The matched rows are reshaped to one row per feature and sample. Where several

rows carry the same feature — duplicates, or several proteins under one gene — their values are reduced to one per sample by the median,

$$v_{fj} = \text{median}\{\, x_{ij} : i \in M_f \,\},$$

over the rows M_f of feature f that have a value in sample j.
Non-numeric and non-finite values become NA. Missing combinations stay in the table and on the sample axis but have no marker. By default the line connects successive observed points. With **Show NA as low**, it visits the display baseline at missing samples without adding a point. No missing value is replaced in the matrix or used as a measurement.

THE ORDER OF THE AXIS

The samples of the selected conditions are relabelled condition_n, numbered within the condition in the order the annotation gives, and laid out condition by condition in the order **Conditions** lists them, which is metadata order. Each sample takes the next integer position; the markers of one feature are then joined in that order.

Use the markers to inspect the individual measurements behind a summary. A lone extreme value deserves a check of that sample and feature; a condition-wide shift may have biological or technical explanations. Missing points show unavailable measurements, not a measured low abundance. The connecting line is a visual guide in the chosen sample order, not a trajectory, interpolation model or evidence of a treatment effect. Use a suitable statistical comparison to assess a difference.

Edge cases

- A condition with one replicate contributes one marker and needs no special handling; there is no minimum number of samples on this page.
- Missing samples remain on the axis. With no observed values there are no measured markers; the optional baseline line must not be mistaken for evidence of a low measured abundance. Zero and negative log2 measurements are valid values and retain their markers.
- When one selection matches several rows, the note above the table says how many were combined and that each value is their per-sample median.
- A selected name that matches no row of the dataset is not drawn. It is named under the figure, in an amber notice box, as Not found in this dataset, so not plotted: …. The notice is shown in the application only.
- The line colours are drawn from tab10, which holds ten, so beyond ten features some lines share a colour.
- At phosphoprotein level collapsed with robust_site_median the values are ratios to each site's own across-sample median, and the axis says log2 Ratio to Site Median instead.

4 The table below the plot

One row per feature and sample, including rows whose Value is NA. The drawing baseline is never written into this table.

Columns in this table

Column	What it holds
Feature	The feature, as labelled in the legend.
Sample	The sample, labelled condition_n as on the axis.
Condition	The condition that sample belongs to.
Value	The plotted value, to four decimals.
SampleCount	How many samples this feature has a value in.
SourceFeatureCount, SourceFeatures	How many rows the selection matched and which — 1 unless a gene maps to several features or the matrix holds duplicates.
Aggregation, Warning	Per-sample median, and a note filled in only when more than one row was combined.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
conditions = ["WT", "GRK"]
fig = compare_prot_line(data, meta, conditions, ["1433B_HUMAN", "1433E_HUMAN"],
                        workflow="Protein", id=False)
```

R

```
conditions <- c("WT", "GRK")
fig <- compare_prot_line(data, meta, conditions, c("1433B_HUMAN", "1433E_HUMAN"),
                         workflow = "Protein", id = FALSE)
```

This call needs no indexed frame: the features are matched against a column, and workflow says which one — ProteinNames, PTM_Collapse_key or Phosphoprotein — in place of the

data level chosen on the page. At phosphosite level the same call takes data3 and meta2 with workflow="Phosphosite", and the features are then PTM_Collapse_key values. The conditions list does the work the page's condition selector does, choosing both which conditions appear and the order they run in along the x axis, and id is **Toggle ID**. Matching rows are reduced to a per-sample median exactly as on the page, but the legend is headed with the workflow name rather than Feature.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the browser scales that render to fit the page. With many samples the labels along the bottom are the first thing to become unreadable, so widen the figure before reducing the font. The table downloads separately as CSV.

Behind this page

Function	What it produces
single_protein_lineplot_plot	The figure
single_protein_lineplot_table	The long table under it
compare_prot_line	The same plot in the package, for a notebook